VisionEval at ImageEval 2026 Shared Tasks: Forced-Choice Reformulation for Hallucination Detection and a Version-Prior Baseline for Cultural Relevance

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Abstract

We describe the VisionEval submissions to ImageEval 2026, covering Task 1b in the English and MSA tracks and Task 2 (CRAI-Bench). For Task 1b the three statements of an item share a stem and exactly one is grounded, so we replace the three independent True/False calls of the released baseline with a single three-way forced choice, read from the logits of the candidate digit tokens. The output then satisfies the one-true constraint by construction, and contrastive instability falls from the baseline 0.2674 to 0.0670 in English and from 0.4277 to 0.1010 in MSA. The organisers' baseline is Qwen2.5-VL-3B prompted one statement at a time, and it identifies the grounded statement about as often as our 7B system does in English (0.9310 against 0.9330), so most of that gap is the output format rather than perception or scale. Task 2 is handled by a six-parameter model over the caption version, which placed first of five teams at Spearman 0.8258; an analysis of what that model reads shows the version field is recoverable from the released instance identifiers, so the result characterises the benchmark more than the system. We also show that the five reported Task 1b columns collapse to one degree of freedom for any system marking one statement true per item, an identity satisfied exactly by all fourteen rows of the final standings.

1 Introduction

Vision-language models assert details that images do not support, and the failure is uneven across languages (Rohrbach et al., 2018; Li et al., 2023; Mousi et al., 2026b). ImageEval 2026 (Abdaljalil et al., 2026) makes this measurable for Arabic. Task 1b pairs an image with three statements differing in one slot, exactly one of which the image supports, and scores systems on *contrastive instability*: how often a system fails to reproduce that one-true pattern. Task 2 (CRAI-Bench) asks how faithfully an AI-generated image realises a Qatari cultural scene described by a caption. We entered Task 1b English, Task 1b MSA, and Task 2.

The released Task 1b baseline asks "True or False?" about each statement in turn and parses the reply. That procedure can return two true statements or none, which is what contrastive instability penalises, and it turns hedged replies into silent errors at the parsing stage. A development run quantifies the cost: prompting this way on the English devtest split produced 631 true labels across 500 items, where the task guarantees exactly 500, so at least 66 items (13%) could not match the one-true pattern regardless of perception (Appendix C). Posing the item as a single three-way choice removes that failure mode by construction.

Constraining the output this way is not, however, a distinguishing idea. Section 5 shows that every system in the final Task 1b standings emitted one true statement per item, so the construction is a precondition for a competitive score rather than a contribution; our submissions placed 7th of 8 and 5th of 6. What this paper offers instead is a set of measurements about the two evaluations: a decomposition separating what the baseline's output format costs from what its smaller model costs (Section 3.3); a proof, confirmed on all fourteen rows of the final standings, that the five reported Task 1b columns carry one free parameter once a system marks one statement true per item (Section 5); and a first-place Task 2 result from six fitted parameters whose backbone reads a released metadata field rather than the image (Section 4.8). Code is available at https://github.com/Md-Faisal-Sheikh/ImageEval.

2 Background and data

Task 1b. An item is an image plus three statements built from a common stem with one substituted slot, for instance a straw that is red with polka

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dots, green with polka dots, or blue with stripes. Data comes from OASIS (Alam et al., 2025), and the contrastive design follows earlier ImageEval probes (Mousi et al., 2026b,a). Systems emit one `true` or `false` row per statement, so 3,000 rows for each track's 1,000 blind test items. Ranking is by contrastive instability, lower being better; the released Qwen2.5-VL-3B baseline scores 0.2674 in English and 0.4277 in MSA.

Task 2 (CRAI-Bench). Each instance pairs a reference photograph of a Qatari scene with an image generated from one of five caption versions, `v1` to `v5`. Systems predict cultural entity accuracy, context, specificity, integrity and hallucination penalty (HP), plus a composite fixed as

$$c = \mathbf{w}^\top \mathbf{d}, \quad \mathbf{w} = (.30, .20, .20, .20, -.10), \quad (1)$$

over the five dimensions $\mathbf{d}$. The splits are small: 120 training instances over 24 reference images, 40 development over 8, and 40 blind test over 8. Ranking is by Spearman ρ on the composite, with mean absolute error alongside. Appendix A gives all split sizes.

3 Task 1b: contrastive hallucination detection

3.1 System

Figure 1(a) shows the pipeline. The trailing "True or False?" tag and its Arabic counterpart are stripped, since they are noise once the item is a choice, and the three statements are numbered inside one prompt (Appendix E) that states the guarantee and asks for a single digit. The instruction is in English in both tracks, with only the statements in the track language.

Rather than generating text, the options are scored from the logits ℓ at the final position (Robinson and Wingate, 2023). With t_0, t_1, t_2 the first token identifiers of `0`, `1` and `2`, the option distribution is

$$p_j = \frac{\exp(\ell_{t_j})}{\sum_{k=0}^{2} \exp(\ell_{t_k})}, \quad j \in \{0, 1, 2\}. \quad (2)$$

This removes the parsing stage and returns an inspectable score per option.

Position preference is handled by rotation. Let R be N cyclic permutations of $\{0, 1, 2\}$ and p^π the option distribution when statement $\pi(j)$ occupies slot j. The score of statement i averages the probability it received wherever it appeared:

$$s_i = \frac{1}{N} \sum_{\pi \in R} p^\pi_{\pi^{-1}(i)}. \quad (3)$$

With $N = 3$ the rotations $(0, 1, 2)$, $(1, 2, 0)$ and $(2, 0, 1)$ place every statement in every slot once. The largest s_i is labelled true and the rest false.

	Inst.↓	Comb.↑	CFHR↓	Q+↑	Q–↑	Rk
English track, 8 teams						
Baseline	0.2674	0.7260	0.2202	0.9310	0.8630	–
VisionEval	**0.0670**	0.9330	0.0000	0.9330	0.9665	7
Best	0.0290	0.9710	0.0000	0.9710	0.9855	1
MSA track, 6 teams						
Baseline	0.4277	0.5660	0.3479	0.8680	0.7900	–
VisionEval	**0.1010**	0.8990	0.0000	0.8990	0.9495	5
Best	0.0360	0.9640	0.0000	0.9640	0.9820	1

Table 1: Official Task 1b test results, 1,000 blind items per track; the baseline is Qwen2.5-VL-3B prompted one statement at a time. Inst. is contrastive instability, Comb. combined accuracy, Rk rank. Only the baseline has nonzero CFHR.

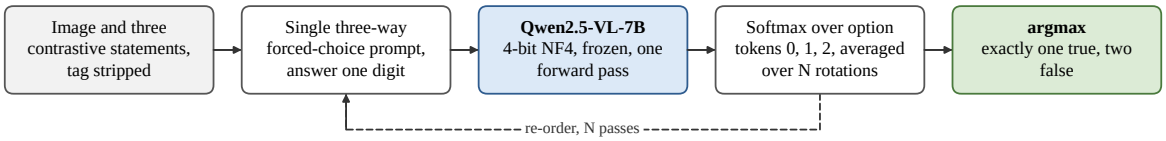
3.2 Experimental setup

Both runs used Qwen2.5-VL-7B-Instruct (Bai et al., 2025) in 4-bit NF4 with double quantisation and bfloat16 compute (Dettmers et al., 2023), through Transformers 5.14.1 (Wolf et al., 2020) on one 16 GB Tesla T4. This part of the work is zero-shot: no labelled Task 1b data was used at any point, and the dev and devtest splits informed only prompt wording. The English run used a single statement ordering ($N = 1$, 1,000 forward passes) and the MSA run three ($N = 3$, 3,000 passes). That difference was an inconsistency between two separately prepared runs rather than a designed contrast, and Section 3.4 treats it as a confound. All 1,000 images of each track downloaded, so both runs cover the full split, and both submissions passed the format checker.

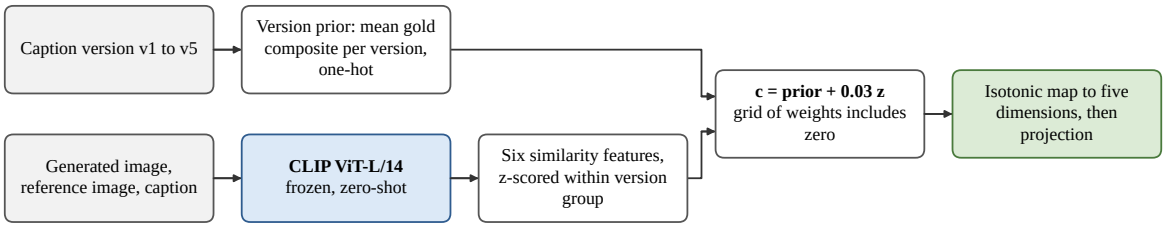
3.3 Results

Table 1 gives the official results. Contrastive instability falls from 0.2674 to 0.0670 in English and from 0.4277 to 0.1010 in MSA, relative reductions of 74.9% and 76.4%, placing 7th of 8 and 5th of 6 teams.

How much of that is the reformulation. The baseline is Qwen2.5-VL-3B prompted one statement at a time, so the comparison spans a change of scale as well as of output format. The two separate,



(a) Task 1b: one three-way forced choice, scored from option-token logits



(b) Task 2: a six-parameter version prior with bounded CLIP tie-breaking

Figure 1: The two submitted systems. In (a) the item is a single choice rather than three independent True/False calls, answered from the logits at the final position of one forward pass; the rotation average was active in MSA (N=3) only. In (b) CLIP features enter as a bounded nudge, z-scored inside each version group, and the composite is recomputed from Equation 1 rather than predicted.

because Q+ measures accuracy on exactly the statements the image supports. In English the baseline scores 0.9310 there and our 7B system 0.9330, a gap a two-proportion test cannot distinguish from zero ($p = 0.86$), while combined accuracy differs by 20.7 points: the 3B baseline finds the grounded statement about as often, and loses by also asserting ungrounded ones, which is what CFHR at 0.2202 against 0.0000 records. In MSA, Q+ differs by 3.1 points ($p = 0.03$) against 33.3 points of combined accuracy, so scale contributes something in Arabic and very little in English. On both tracks the output format, not perception and not model size, accounts for most of the reduction.

This is a decomposition, not a controlled ablation: Q+ is a proxy for perception rather than a measurement of it. The clean experiment remains the one in Appendix C, the same 7B checkpoint prompted per statement.

3.4 Analysis

Confidence under-states accuracy. The mean of $\max_j s_j$ is 0.901 in English against a realised item accuracy of 0.9330, and 0.813 in MSA against 0.8990, with the count of items below 0.5 pointing the same way (102 in MSA against 101 errors, 45 in English against 67). This compares two summary statistics and is not a calibration analysis: per-item gold for the blind splits is not released, so no reliability diagram or expected calibration error is computable here, and no claim about calibration in the formal sense (Guo et al., 2017) is intended.

Position preference and the two tracks. The chosen index is close to uniform in both tracks: English at $N = 1$ gives $(315, 356, 329)$ and MSA at $N = 3$ gives $(330, 346, 324)$, so $\chi^2 = 2.61$ and $\chi^2 = 0.78$ on two degrees of freedom, neither departing significantly from uniform. Under an explicit forced choice, then, position preference is small before any averaging is applied. The two tracks differ in item accuracy by 3.4 points ($p = 0.006$, two-proportion test), but they also differ in two respects at once, language and rotation count, so we attribute that gap to neither. A flat index histogram is consistent with position preference being absent, yet rotation also averages three prompts, which affects more than position. One English run with $N = 3$ would separate the two, and until that run exists the gap is reported without explanation.

4 Task 2: cultural relevance assessment

4.1 Overview

The Task 2 system runs in four stages. **(1)** The caption version selects a prior value, the mean gold composite of that version. **(2)** A frozen CLIP encoder turns the generated image, the reference image and the caption into six similarity features, reduced to one signal and standardised inside each version group. **(3)** Prior and signal combine into a predicted composite. **(4)** That composite is expanded into the five CRAI dimensions by monotone maps and projected so the official formula holds. Figure 1(b) shows the flow; the subsections

Stage	Fitted on labels	Applied at test
1 Prior	5 version means	lookup by version
2 CLIP	nothing (frozen)	6 features, 1 signal
3 Blend	1 weight β	$\hat{c} = \mu_v + \beta z$
4 Expand	5 isotonic maps	project to $c^\star$

Table 2: Task 2 training and inference pipelines. Six scalar parameters and five monotone maps are estimated from the 160 labelled train and development instances; nothing else is learned.

take the stages in order.

Unlike the Task 1b system, this pipeline is *not* zero-shot. Stages 1, 3 and 4 all depend on quantities estimated from labelled task data: five version means, one blend weight, and five monotone maps. Only the CLIP encoder is used without adaptation. Table 2 separates what is fitted at training time from what is computed at inference.

4.2 Stage 1: the version prior

Three measurements on the released training labels motivate the design, and each contradicts the task description. First, the versions are described as running from most to least culturally specific, but the mean gold composite per version is 0.8009, 0.5356, 0.2252, 0.5428 and 0.0871 for `v1` to `v5`, so `v4` sits 0.32 above `v3` and an ordinal encoding would impose an ordering the labels do not have. The version is therefore one-hot. Second, hallucination penalty is exactly zero on 96.7% of training rows, making its contribution to Equation 1 nearly constant. Third, the other four dimensions inter-correlate at Spearman 0.81 to 0.97 and one latent factor accounts for 92.6% of their variance, so predicting one number and expanding it costs little while fitting far fewer parameters than five separate regressors on 120 rows. The prior μ_v is the mean gold composite of version v, estimated on the 120 training and 40 development instances together (Appendix D.1).

4.3 Stage 2: CLIP tie-breaking signal

The prior assigns all eight instances of a version the same value, leaving five tie groups. On the development split a prior with perfectly ordered ties would score $\rho = 0.8527$ against 0.6823 for the prior alone, so roughly half the attainable signal sits inside those groups.

A frozen CLIP ViT-L/14 encoder (Radford et al., 2021) embeds the generated image, the reference image and the caption, yielding six similarity features (Appendix D.3). The most useful measure the margin between mean similarity to five Qatari-specific text probes and five generic ones, which are separated by 0.3135. Features correlating below 0.10 in absolute value with the within-version gold residual are dropped; the rest are combined by sign into one signal and standardised within each version group to give z. Standardising within the group is what confines this stage to reordering instances inside a version and prevents it from disturbing the ordering between versions.

4.4 Stage 3: blending

The predicted composite is

$$\hat{c} = \mu_v + \beta z, \tag{4}$$

with β selected over a grid that includes zero. That zero is a safeguard: if the visual signal is noise, the search returns $\beta = 0$ and the system falls back to the prior without intervention. The submitted run used $\beta = 0.03$.

4.5 Stage 4: expansion and projection

Each dimension is recovered from $\hat{c}$ by an isotonic regression g_m fitted on the same labelled pool and clipped to $[0, 1]$ (Pedregosa et al., 2011), since the relation is monotone but visibly nonlinear. The resulting vector $\mathbf{d}_0 = (g_1(\hat{c}), \ldots, g_5(\hat{c}))$ need not satisfy Equation 1, so it is moved the shortest distance along $\mathbf{w}$ until it does, clipped to $[0, 1]$ after each of six steps (Appendix E). Writing $\mathbf{d}$ for the projected vector, the *reported* composite is

$$c^\star = \mathbf{w}^\top \mathbf{d}, \tag{5}$$

which is distinct from the intermediate prediction $\hat{c}$ of Equation 4. Reporting $c^\star$ rather than $\hat{c}$ makes the submitted file consistent with the official formula by construction.

4.6 Experimental setup

The prior, the isotonic maps and the blend weight were fitted on the combined train and development gold, 160 instances over 32 reference images. The weight β was chosen by six-fold cross-validation grouped by `image_id`, which matters because the five versions of one reference image are near-duplicates and ungrouped folds leak. CLIP ran frozen in float16 on the same T4. The submission passed the organisers' format checker. Appendix E

System	Spearman ↑	MAE ↓
GPT-4o judge (baseline)	0.5186	0.2357
VisionEval (1st of 5)	**0.8258**	0.1469
Best competing	0.8078	0.1512
Lowest MAE submitted	0.8040	**0.1415**

Table 3: Official Task 2 test results, 40 blind instances over 8 reference images. The organisers' baseline is a GPT-4o judge.

gives the remaining hyperparameters. Every number reported below comes from the official leaderboard or from the recorded output of the submitted runs, and no post-submission experiment is included.

4.7 Results

Table 3 gives the official result: first of five teams at $\rho = 0.8258$, 0.018 ahead of the runner-up and 0.307 above the organisers' GPT-4o judge baseline. One lower-ranked submission reached a better MAE (0.1415 against 0.1469) with a lower ρ, which follows from reporting $c^\star$ after projection, a step that favours ordering over pointwise error.

The test score is 0.138 above the corresponding development score, and the development-phase standing was behind the team finished ahead of on test (0.6413 against 0.7076). Feature-rich alternatives did not hold up (Appendix D.2): grouped cross-validation rose from 0.7106 to 0.8094 as features were added while the development score fell from 0.6650 to 0.4998. One family needs a note: `gold_llm.tsv` is a file *shipped with the task release* holding GPT-4 judge scores, so no commercial API was called by this system and the file served only as an input feature during development. Those scores correlate with gold at $\rho = 0.645$ on train but 0.249 on development, and adding them produced the worst development result of any configuration tried.

4.8 What the version prior exploits

The obvious question about a six-parameter model that wins a visual task is whether it looks at the images at all. It largely does not. The caption version is not inferred: the released `captions.tsv` gives each instance an id whose final underscored field is the version, so the backbone is a lookup on a field the benchmark distributes with the blind test split, and the CLIP stage moves predictions only within a version group.

This makes the first-place result a statement about CRAI-Bench rather than about the system: the generating caption predicts how well an image realises the scene and is released alongside the blind instances, so a system can score well while ignoring the generated image. Future editions could withhold the version field at test time, or score only within-version orderings, where the genuinely visual signal sits and where roughly half the attainable development signal remains unclaimed.

5 What the Task 1b scoreboard measures

Let p be the fraction of items on which a system selects the grounded statement. Any system marking exactly one statement true per item satisfies three identities: accuracy on grounded statements is p, since there is one per item; accuracy on ungrounded statements is $(1 + p)/2$, since a correct selection labels both correctly while an incorrect one mislabels one of the two; and combined item accuracy is again p, because all three statements are correct exactly when the selection is. Contrastive instability is therefore $1 - p$.

All fourteen rows of the final standings satisfy these identities to the four published decimals, and CFHR is uniformly zero. Every submitted system emits one true statement per item, so the five reported columns are five views of a single number. The released baseline is the counterexample that fixes the scope of the claim: it does not mark one statement true per item, and it satisfies none of the identities, with $1 -$ Inst at 0.7326 against a combined accuracy of 0.7260 in English and a nonzero CFHR in both tracks. Two consequences follow. The construction of Section 3 is a precondition for a competitive score rather than a differentiator, which is why we do not present it as our main contribution. And among submitted systems the metric no longer separates the inconsistency it was designed to detect (Mousi et al., 2026b) from plain accuracy; recovering it needs either unconstrained systems or a statistic that is not a function of p alone.

6 Conclusion

Three measurements survive. Against a 3B baseline that finds the grounded statement about as often as our 7B system, almost all of the Task 1b gap is output format rather than perception. The standings carry one degree of freedom: the five reported columns describe a single quantity. And the first-place Task 2 result came from a lookup.

Limitations

The headline Task 1b comparison is not a controlled ablation. The baseline is Qwen2.5-VL-3B and our systems are 7B, so scale and output format differ together. The Q+ decomposition in Section 3.3 bounds the scale contribution using accuracy on the grounded statements, but Q+ is a proxy for perception rather than a measurement of it. The missing experiment is one run: the same 7B checkpoint prompted per statement on the English devtest split, scored by the label-count argument of Appendix C, which needs no gold labels.

The two Task 1b submissions differ in statement language and in rotation count at the same time, an inconsistency between separately prepared runs rather than a designed contrast. No effect of rotation averaging, and no language effect, can be estimated from this pair. One English run at $N = 3$ would resolve both.

The confidence figures in Section 3.4 compare two summary statistics. Per-item gold labels for the blind splits are not released, so nothing we report establishes calibration in the formal sense, and the reported under-confidence should be read only as an aggregate discrepancy.

Task 2 conclusions rest on 24 training and 8 development reference images, small enough that the development split is itself a noisy selection criterion, and the test result comes from 8 blind reference images. Only one Task 2 configuration was submitted, so the contribution of the CLIP stage on the test split is unmeasured; its value is established only on development. The cultural probes are written in English and name Qatari referents, so that component should not be assumed to transfer to other Arab settings without re-checking probe separation. Finally, the degree-of-freedom result in Section 5 describes the fourteen rows of this year's standings; it is a property of systems that adopt the one-true form, not a theorem about the metric under all submissions.

Ethics Statement

This work uses only data released by the ImageEval 2026 organisers and two open-weight checkpoints, and complies with the terms of the shared task. No human subjects were involved and no annotation was collected. The evaluation concerns depictions of Qatari cultural scenes; the cultural probes used for tie-breaking were written in English by researchers outside that community, and they encode an outsider's list of salient referents. They should be treated as a convenience feature rather than as a considered representation of the culture, and any reuse for cultural assessment should involve community review. Section 4.8 reports that strong performance on CRAI-Bench is reachable without attending to the generated image; presenting that openly matters, because a benchmark result read as evidence of cultural understanding, when the system reads a metadata field, would misrepresent what the model can do. Compute for all reported runs came from the free Kaggle notebook tier, with a total budget of a few GPU hours.

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A Data and splits

Task	Split	Items	Rows	Images
1b English	devtest	500	1,500	500
	test	1,000	3,000	1,000
1b MSA	test	1,000	3,000	1,000
2 (CRAI-Bench)	train	120	120	24
	dev	40	40	8
	test	40	40	8

Table 4: Split sizes. For Task 1b, rows is items $\times$ 3 statements; for Task 2, images is the number of distinct reference photographs, five caption versions each.

B Submitted run statistics

	English	MSA
Rotations N	1	3
Forward passes	1,000	3,000
Images retrieved	1,000/1,000	1,000/1,000
Submitted rows	3,000	3,000
Mean $\max_j s_j$	0.901	0.813
Items $\max_j s_j < 0.5$	45	102
Chosen index	(315,356,329)	(330,346,324)
χ^2 against uniform	2.61	0.78
Format check	passed	passed

Table 5: Configuration and recorded output of the two submitted Task 1b runs, both on Qwen2.5-VL-7B-Instruct in 4-bit NF4 on one Tesla T4.

C Development-phase Task 1b experiment

Before adopting the forced choice, the per-statement True/False procedure was run on the English devtest split with a smaller checkpoint, to observe the output distribution when the one-true guarantee is not enforced (Table 6). Parsing was not the bottleneck: every reply was readable and none was defaulted. The label counts are the problem. The split guarantees exactly 500 true statements and the run produced 631, an excess of 131, so at least 66 of the 500 items carry two or more true labels. That bound holds however the excess is distributed and requires no gold labels, which is what makes the same measurement cheap to repeat at 7B.

Because this run used a 3B checkpoint and the submitted runs used 7B, the table below bounds what the output format costs at one scale. It does not isolate the format from the model, and we do not use it for that purpose.

Setting	Value
Split	English devtest (500 items)
Checkpoint	Qwen2.5-VL-3B
Procedure	one True/False call each
Judgements	1,500
Unparseable replies	0
Predicted true / false	631 / 869
Expected true	500
Items off the one-true pattern	$\geq$ 66 (13%)

Table 6: Development-phase run with independent True/False scoring. The devtest split is blind, so no accuracy is available; the inconsistency is measurable without labels.

D Task 2 model internals

D.1 Version prior

Version	Mean gold composite	Rank by value
`v1`	0.8009	1
`v2`	0.5356	3
`v3`	0.2252	4
`v4`	0.5428	2
`v5`	0.0871	5

Table 7: Version prior, fitted on the 160 combined train and development instances. `v4` outranks both `v3` and `v2`, so the published v1 to v5 ordering does not hold in the labels.

D.2 Development-phase system comparison

System	CV ρ	Dev ρ	Dev MAE
Version prior (5 numbers)	n/a	0.6823	0.2054
Version one-hot, Ridge	0.7106	0.6650	0.2069
+ caption lexicon	0.7695	0.5643	0.2227
+ pixel statistics	0.7642	0.5979	0.2368
+ `gold_llm` features	0.8094	0.4998	0.2735
Shipped `gold_llm` scores	n/a	0.2493	0.3761
Prior + perfect tie-break	n/a	0.8527	0.2052

Table 8: Development-phase comparison for Task 2. The version prior here is fitted on train only, so its development score (0.6823) differs from the train-plus-dev fit used in the submitted system, which scores 0.6650 at $\beta = 0$. The `gold_llm` rows use GPT-4 judge scores distributed with the task release; no commercial API was called by this system.

Selecting the blend weight. Any nonzero weight up to 0.08 improves on the prior and the curve is level across that range (Table 9), which is how a bounded nudge should behave. The submitted run used $\beta = 0.03$, the cross-validation maximum,

β	.00	**.01**	.02	**.03**	.05	.08	.12	.18	.25
CV ρ	.802	.808	.810	**.810**	.809	.805	.787	.737	.665
Dev ρ	.665	**.699**	.689	.688	.686	.692	.679	.644	.596

Table 9: Blend weight selection, a development-phase experiment. Cross-validation is grouped by `image_id` over the 120 training instances; the development column is the held-out 40. The criteria peak at different weights: .03 was submitted, .01 is the development optimum.

which is not what the selection procedure was supposed to do: the notebook text specifies reading the weight from the development column, which selects $\beta = 0.01$ at $\rho = 0.6993$, but the executed loop compared cross-validation scores and returned 0.03 at $\rho = 0.6879$. The cost is 0.011 development ρ and both values sit above the $\beta = 0$ row, so the submitted file was not harmed. It is reported because a gap between stated protocol and executed code is easy to leave undocumented.

D.3 CLIP feature screening

Feature	ρ with residual	Kept
`cult_delta`	$+0.1739$	yes
`cult_margin`	$+0.1714$	yes
`follow_gap`	-0.1287	yes
`sim_gen_ref`	$+0.1279$	yes
`cult_max`	$+0.1178$	yes
`sim_gen_cap`	-0.0303	no

Table 10: CLIP feature screening on the 120 training instances, retention threshold $|\rho| \geq 0.10$, correlations against the within-version gold residual. `cult_margin` is mean similarity to the five cultural probes minus mean similarity to the five generic probes; `cult_delta` subtracts the same quantity measured on the reference image; `follow_gap` is the caption similarity of the generated image minus that of the reference; `cult_max` is the maximum similarity to any cultural probe.

E Replication details

Task 1b. `Qwen/Qwen2.5-VL-7B-Instruct`; bitsandbytes 4-bit NF4 with double quantisation and bfloat16 compute, disabled automatically below compute capability 7.5; processor `max_pixels` $= 512 \times 28 \times 28$; greedy option scoring with no sampling; rotations $N = 1$ (English, Codabench competition 17051) and $N = 3$ (MSA, competition 17050); per-item checkpointing with resume; PyTorch 2.10.0 with CUDA 12.8; data from `QCRI/AynVQA-ArabicNLP26`, files `task1b/test_en.jsonl` and `task1b/test_msa.jsonl`. Runtime is roughly one to one and a half hours per 1,000 items at $N = 3$ on one T4. The prompt is:

```
Look at the image.  Exactly
ONE of the following three
statements is true of this
image; the other two are
plausible but NOT supported
by it.

0.  {s0}
1.  {s1}
2.  {s2}

Judge only by what the image
actually shows.  Which statement
is the true one?  Answer with a
single digit, 0, 1 or 2.
```

Task 2. `openai/clip-vit-large-patch14`, frozen, float16, batch size 16; isotonic regression from scikit-learn with `out_of_bounds="clip"` and range $[0, 1]$; the projection step $\mathbf{d} \leftarrow \mathrm{clip}_{[0,1]}\big(\mathbf{d} + (\hat{c} - \mathbf{w}^\top \mathbf{d})\,\mathbf{w}/(\mathbf{w}^\top \mathbf{w})\big)$ iterated six times from $\mathbf{d}_0$, after which $c^\star = \mathbf{w}^\top \mathbf{d}$ is reported; blend grid $\{0, .01, .02, .03, .05, .08, .12, .18, .25\}$; six-fold `GroupKFold` on `image_id`; random seed 42. The caption version is taken from the `version` column of `captions.tsv` when present and otherwise from the final underscored field of the instance id, both of which are available for the blind test split. Predicted cultural specificity values are left continuous rather than snapped to the five tiers named in the task description, since the gold labels are themselves continuous (45 distinct values in train) and the official checker treats non-tier values as a warning rather than an error.